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SUPERSEDING
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# MILITARY SPECIFICATION

## RECORDER, MAGNETIC DISTORTION, RO-32/ASQ

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

## 1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide a dual recording milliammeter capable of recording energy in the frequency range of 0 to 3 cps on suitable paper when energy is applied from either or both of two separate sources.

1.2 Classification - The Recorder, Magnetic Distortion, RO-32/ASQ shall be of one type and shall consist of the following:

| Item | Type Designation | Max. Allow. Size (Inches) W | H/L | D | Max. Allow. Wt. Lbs. | Appl. Para. |
|---|---|---|---|---|---|---|
| Recorder, Magnetic Distortion | RO-32/ASQ | 5-3/4 | 7-3/8 | 6-1/2 | 8.8 | 3.5 |

## 2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-E-4682 | Electron Tubes and Transistors, Choice and Application of |
| MIL-W-5088 | Wiring, Aircraft, Installation of |
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification for |
| MIL-T-5422 | Testing; Environmental Aircraft Electronic Equipment |
| MIL-I-6181 | Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment |
| MIL-C-6781 | Control Panel; Aircraft Equipment, Rack or Console Mounted |
| MIL-P-7788 | Plate, Plastic, Cockpit and Interior Controls Lighting |

FSC 6655



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| | |
|---|---|
| MIL-P-17555 | Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts |
| MIL-T-18303 | Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment; Format for |
| MIL-N-18307 | Nomenclature and Nameplates for Airborne Electronic and Associated Equipment |
| MIL-F-25173 | Fastener, Control Panel, Aircraft Equipment |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-704 | Electric Power, Aircraft, Characteristics and Utilization of |
| MS 25245 | Connections - Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, and/or 115/200-Volt Three Phase AC, or combinations |

PUBLICATIONS

Bureau of Naval Weapons

| | |
|---|---|
| EI-152 | Avionics Installation Instruction for Magnetic Distortion Recorder RO-32/ASQ |
| SAR-300 | Material Changes and Material Bulletins; Preparation of |

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400,

(2) Nonrepairable subassemblies, as outlined in specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable A mean-time-to failure of more than 5000 hours should be the goal of each.



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(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction -

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class I equipment, with the following modifications:

3.2.2.1 Temperature - The equipment shall operate satisfactorily over the temperature range of minus 30° to plus 55°C.

3.2.2.2 Altitude - The equipment shall operate satisfactorily at all altitudes from sea level to 10,000 feet. No permanent damage shall result when the equipment is transported to any altitude up to 50,000 feet while in an operating or non-operating condition. Satisfactory operation shall resume immediately when pressure equivalent to 10,000 feet altitude is returned.

3.2.3 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment.

| | |
|---|---|
| Temperature | Room ambient 25°C ±15°C |
| Altitude | Normal ground |
| Vibration | None |
| Humidity | Room ambient up to 90% relative humidity |
| Input power voltage | 115 ±1.0 V AC and/or 27.5 ±0.5 V DC |

3.2.4 Total Weight - The total weight of the equipment, excluding cables and including chart paper, shall be a minimum consistent with good design and shall not exceed 8.8 pounds.

3.2.5 Operating Life -

3.2.5.1 Operational Stability - The equipment shall operate with optimum performance for 200 hours, continuously or intermittently, without necessity for readjustment of any controls which are inaccessible during flight.

3.2.5.2 Reliable Operating Life - The equipment shall have a reliable operating life of at least 500 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.5.3 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.



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3.2.6 Cabling and Connections -

3.2.6.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.6.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.7 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.8 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-STD-704 and shall give specified performance when provided with power from the following sources with characteristics as defined in Specification MIL-STD-704. The power required shall not exceed the amounts shown:

115 Volts, 1Ø AC Power, Category "C" 50 V.A.

28 Volts, DC Power Category "B" 1 amp.

3.2.8.1 AC Frequency - The equipment shall not be damaged by power supply frequencies below the minimum specified for a period of 30 minutes, and shall automatically resume normal operation when the frequency returns within limits; however, there are no requirements on equipment performance at frequencies below the lower established operating limit.

3.2.9 Input Power Connection - Input power connections shall be in accordance with MS 25245, except that the inverter starting relay circuit is not required.

3.2.10 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuzes and other protective devices shall be held to a minimum.

3.2.10.1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.2.11. Moisture and Fungus Resistant Treatment - Equipments shall be fungus-proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that over-all spraying of the equipment is not necessary.

3.2.12 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307. See 6.4.

3.2.13 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3 2.14 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.



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3.2.15 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3 Performance - The performance of the Recorder RO-32/ASQ shall meet the requirements of this specification.

3.3.1 Warm-Up Time - No warm-up time for the equipment shall be required under standard conditions. The warm-up time shall not exceed 15 minutes at extreme service conditions.

3.3.2 Frequency Characteristics - The equipment shall record satisfactorily over the input frequency range of 0 to 3 cps (upper half power point at 4 ± 1 cps).

3.3.3 Recording - The equipment shall produce a continuous graphic recording of input signals within the frequency range specified herein.

3.3.4 Range Scale - Recording shall be on a 4-inch curvilinear chart (50 minor chart divisions) with center zero. Time markings shall be a light line every 1/2 inch with a heavy line every three (3) inches. In addition, markers shall be provided by which time increments down to one second may be estimated at a paper speed of three (3) inches per minute.

3.3.5 Recorder Sensitivity - Full-scale deflection of the recording pen from the zero center shall result from an input of 0.5 milliampere of either a positive or negative polarity signal over the range of 0 to 3 cps. The recording shall, throughout the range scale, be a true representation of the input to an accuracy of one minor chart division over the range of 0 to 3 cps under all service conditions.

3.3.6 Paper Speed - The paper speed shall be 3/4 inch, 3 inches, and 12 inches per minute. Paper speed shall be accurate to within ±3% at a line frequency of 400 cycles per second. The paper speed shall be directly proportional to line frequency changes from 380 to 420 cycles.

3.3.7 Provision for Use of Chart Paper - A take-up spool shall be provided for used chart paper, and provisions shall be incorporated for allowing examination of this chart when desired. It shall be possible to write on the recording chart concurrently with recording in order to identify signals.

3.3.8 Reloading - Reloading of the recording chart shall be possible with the recorder installed in its mounting rack.

3.3.9 Platen Lighting - The chart platen shall be illuminated with white light so that there is a relatively uniform brightness over the entire viewing area. The maximum brightness shall be not greater than 7 times the dimmest areas. The dimmest areas shall be not less than 0.50 foot lambert and the brightest areas shall be not more than 7.5 foot lamberts when operated at full brilliancy.

3.4 Detail Requirements - The Recorder, Magnetic Distortion RO-32/ASQ, shall meet the following requirements:

3.4.1 Function - The Recorder, Magnetic Distortion RO-32/ASQ, shall provide two closed loop servo systems capable of driving 2 two-phase, 400-cycle servo motors. A potentiometer shall be connected to each pen gear train to feed angular position data back to the servo amplifier. The input of each recording channel shall be designed with an input impedance of 15,000 ±1500 ohms while operating, to provide for the recorder to be used with Detecting Set AN/ASQ-8 or AN/ASQ-10. An electrical signal of 0.5 milliampere shall cause a full-scale deflection of each recording pen from zero center according to the positive or negative polarity of the signal in a frequency range of 0 to 3 cps.



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3.4.2 Form Factor and Mounting - The recorder shall conform in its proportions with the applicable requirements of Specification MIL-C-6781 and shall not be greater than 5-3/4 inches wide by 6-1/2 inches deep by 7-3/8 inches long as shown in the applicable drawing of EI-352.

3.4.3 Contents of Unit - The recorder shall contain the following:

3.4.3.1 Pen Drive Motors - A two-phase, 400-cycle AC servo motor shall be provided to drive each of the recording pens. The fixed field shall be wound for 115 volts. The variable field shall be wound for 40 volts center tapped. The output shaft shall contain a 13-tooth, 120-pitch, 20° pressure angle gear.

3.4.3.2 Electronic Control Amplifier - An electronic control amplifier shall be provided for each of the two servo loops. The amplifier shall have plug-in connections to fit a Type K-10S printed circuit connector. The electrical connections shall be as follows:

| Pin Letter | Function |
|---|---|
| A | Output |
| B | Plus 28 Volt DC Supply |
| C | Not Used |
| D | Input |
| E | Input Center Tapped |
| F | Input |
| H | Gain Control |
| J | Ground |
| K | Not Used |
| L | Output |

3.4.3.3 Servo Potentiometers - A center-tapped servo potentiometer shall be provided for each servo channel.

3.4.3.4 Calibration Control - A calibration control shall be provided in the input circuit of each channel.

3.4.3.5 Zero Adjustment - A zero adjusting control shall be provided to mechanically adjust the position of the servo potentiometer.

3.4.3.6 Chart Drive Motor - A chart drive motor shall be provided to drive the chart at a constant speed and to drive paper take-up spool.

3.4.3.7 Recording Paper Chart - A recording paper chart shall be provided.

3.4.3.8 Control Panel Assembly - The control panel shall be recessed between the two-side mounting rails and shall contain all operating controls.

3.4.3.9 Chart Platen Illumination Adjustment - A control shall be provided on the control panel assembly to adjust the chart platen illumination. This control shall be independent of the test switch.

3.4.3.10 Paper Take-Up - Paper take-up mechanism shall be provided to properly drive the paper take-up spool through a slipping clutch.

3.4.4 Controls -

3.4.4.1 Panel Controls - The front panel assembly shall incorporate the following controls:

(1) An eight-position rotary power and test switch.

(2) Illumination intensity control.



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3.4.4.2 Identification Lettering - The lettering on the control panel assembly shall identify the controls as follows:

3.4.4.2.1 Test Switch - The test switch shall be labeled "TEST" and shall have positions marked as follows:

(1) "OFF" Equipment off.

(2) "OPR" Equipment is on in normal operating condition.

(3) "R" The black pen is in normal operating condition while the red pen is at center zero.

(4) "+" The red pen is deflected four major divisions to the right while the black pen is in normal operation.

(5) "-" The red pen deflects four major divisions to the left while the black pen is in normal operation.

(6) "B" The black pen is in zero position while the red pen is in normal position.

(7) "+" The black pen is deflected four major divisions to the right while the red pen is in normal operation.

(8) "-" The black pen is deflected four major divisions to the left and the red pen is in normal operation.

3.4.4.2.2 Platen Illumination Control - A rheostat control labeled "ILLUM" with both an OFF position and an arrow shall be provided to control the chart illumination.

3.4.4.3 Interior Controls - The following controls shall be accessible for inflight adjustment.

3.4.4.3.1 Zero Centering Adjustment - A control for adjusting each of the recording pens to zero center shall be incorporated and it shall be easily accessible with a screwdriver during flight. Clockwise rotation of this control shall move the zero position of the recording pen to the right.

3.4.4.3.2 Calibration Potentiometer - Calibration control potentiometers (one for each channel) shall be provided to control the recorder sensitivity response to maintain the recorder input sensitivity as specified in 3.3.5.

3.4.5 Electrical Connections - All electrical connectors shall enter the recorder from the back in accordance with Drawing EI-352-1.

3.4.6 Recording Pens - The two recording pens shall be ball-point type and shall contain black and red inks. The lower pen shall be black and the upper pen red.

3.4.7 Visible Record - A minimum of 4.5 inches of previous recording record shall be visible at all times.

3.4.8 Control Panel Assembly - The control panel assembly shall include a plastic lighting plate conforming to Specification MIL-P-7788 to provide illumination of the identification lettering for the front panel controls.

3.4.9 Access Door - A transparent door shall be included in the recorder to complete the enclosure. The door shall be capable of being stowed within the unit when not required.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment. It may not be resubmitted as the production sample model.



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4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when required by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - After completion of the production sample tests the equipment shall be returned to the contractor for any necessary rework. The equipment shall then be resubmitted for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.



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4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

| | |
|---|---|
| Temperature | Ambient room |
| Humidity | Ambient room |
| Vibration (10 minutes) | Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's |

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

| Quantity of Equipments Offered for Acceptance | Quantity to be Tested |
|---|---|
| First 10 | 1 |
| Next 50 | 1 |
| Next 75 | 1 |
| Next 100 | 1 |
| | 1 for each additional 200 or fraction thereof |

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

(3) Manufacturing reliability test specified in 4.4.1.3, except that the test duration shall be 30 hours.



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4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.



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4.5 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

| Quantity of Equipments Offered for Acceptance | Quantity to be Tested |
|---|---|
| First 25 | 1 |
| Next 175 | 1 |
| Next 300 | 1 |
| | 1 for each additional 500 or fraction thereof |

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

| | |
|---|---|
| Temperature | Normal room |
| Altitude | Normal ground (0 - 5000 ft.) |
| Humidity | Room ambient |
| A.C. Voltage | 115 ± 5 volts (at lowest applicable frequency) |
| D.C. Voltage | 27.0 ± 2.0 volts |

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.



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4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.9 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes, and Material Bulletins shall be prepared in accordance with SAR-300.

## 5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.



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## 6. NOTES

6.1 Use - The equipment shall be designed for use in Naval aircraft to carry out specific military missions.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±0.5 V, DC and 115 ±1 V, AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specifications.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 The parentheses ( * ), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.12. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.5 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.



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